Name of Student: Minisha Singh	**PRN: 25070122292**
Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 22	
TITLE : JDBC PreparedStatement, and ResultSet	

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

Learning Objectives

To implement PreparedStatement and ResultSet for secure query execution and data retrieval.

Exercise

1. Create a login application using PreparedStatement to check username and password.

Code:

```
import java.sql.*;
import java.util.Scanner;

public class Login {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "mini@12";

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter username: ");
        String username = sc.nextLine();

        System.out.print("Enter password: ");
        String pass = sc.nextLine();
```

```
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);

            String query = "SELECT * FROM login WHERE username = ? AND
password = ?";

            PreparedStatement ps = con.prepareStatement(query);

            ps.setString(1, username);
            ps.setString(2, pass);

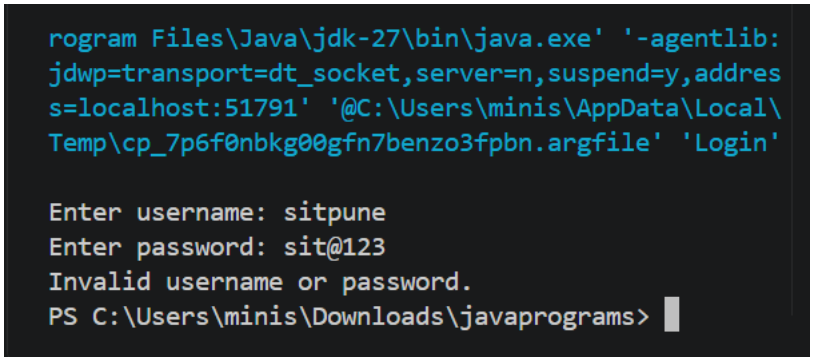
            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                System.out.println("Login successful.");
            } else {
                System.out.println("Invalid username or password.");
            }

            con.close();
            sc.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Output:



```
'c:\Users\minis\Downloads\javaprograms'; & 'C:\Pro
gram Files\Java\jdk-27\bin\java.exe' '-agentlib:jd
wp=transport=dt_socket,server=n,suspend=y,address=
localhost:59535' '@C:\Users\minis\AppData\Local\Te
mp\cp_7p6f0nbkg00gfn7benzo3fpbn.argfile' 'Login'
Enter username: admin
Enter password: 1234
Login successful.
PS C:\Users\minis\Downloads\javaprograms>
```

2. Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

Code:

```
import java.sql.*;
import java.util.Scanner;

public class HospitalLogin {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "mini@12";

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter role (doctor/nurse): ");
        String role = sc.nextLine();

        System.out.print("Enter login ID: ");
        String loginId = sc.nextLine();

        System.out.print("Enter password: ");
        String pass = sc.nextLine();

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);
```

```
        String query = "SELECT * FROM hospital_staff WHERE role = ? AND
login_id = ? AND password = ?";

        PreparedStatement ps = con.prepareStatement(query);

        ps.setString(1, role);
        ps.setString(2, loginId);
        ps.setString(3, pass);

        ResultSet rs = ps.executeQuery();

        if (rs.next()) {
            System.out.println("Login successful. Access granted.");
        } else {
            System.out.println("Invalid credentials. Access denied.");
        }

        con.close();
        sc.close();

      } catch (Exception e) {
        System.out.println(e);
      }
   }
}
```

Output:

```
jdwp=transport=dt_socket,server=n,suspend=y,addres
s=localhost:65401' '@C:\Users\minis\AppData\Local\
Temp\cp_7p6f0nbkg00gfn7benzo3fpbn.argfile' 'Hospit
alLogin'
Enter role (doctor/nurse): doctor
Enter login ID: doc101
Enter password: 1234
Login successful. Access granted.
PS C:\Users\minis\Downloads\javaprograms>
```

Conclusion

In this practical, we learned how to use JDBC classes such as PreparedStatement and ResultSet for database operations. We created login applications to verify user credentials securely using database records. The programs successfully authenticated valid usernames, passwords, and hospital staff details and displayed appropriate access messages. This helped us understand secure query execution and retrieving results from a MySQL database using JDBC.